Speech recognition on UCloud: accuracy and energy

Natanael Fejes, SDU Sønderborg · measured 30 September 2026 on UCloud (1× NVIDIA B200 per run) · public test sets only

1. Danish: models offered in UCloud Transcriber vs edda-v0.1 (word error rate, %, lower is better)

Model	CoRal conversation	CoRal read-aloud	FLEURS	FTSpeech	Energy, Wh per audio hour*
edda-v0.1 (open, Apache-2.0)	**16.0**	**10.5**	**7.6**	**6.0**	**5.9 / 7.7**
whisper-large-v3 (Transcriber default)	48.7	26.9	12.8	16.5	23.4 / 31.5
whisper-large-v3-turbo	61.8	31.3	13.0	18.5	4.8 / 10.1
whisper-medium	64.6	37.1	21.1	29.1	18.8 / 25.4
whisper-small	81.5	49.5	33.5	36.4	7.8 / 13.8

* first value measured on FLEURS + FTSpeech, second on CoRal. edda's WER reproduces its published model card exactly. Parakeet (also in Transcriber) was not re-measured; its published mean WER on the same benchmark is 30.7%.

2. Accented English (word error rate, %, and energy)

Model	EdAcc (25 accent groups)	FLEURS (clean, read)	Energy on EdAcc, Wh per audio hour
whisper-large-v3-turbo	**19.0**	**6.1**	**3.9**
whisper-large-v3 (default)	21.1	5.4	25.0
whisper-medium	31.2	6.0	20.9
whisper-small	38.0	7.6	10.6

On very short meeting clips (AMI), the default model sometimes loops ("Yeah. Yeah. Yeah…"), costing about 108 Wh per audio hour instead of about 9 for turbo, for worse output. Accuracy benchmarks alone do not show this.

3. Suggestion for Transcriber

- Add **edda-v0.1** as the Danish option: 1.7–3× fewer errors than the default (3× on conversational speech) at about a quarter of the energy.
- Make **large-v3-turbo** the English default: more accurate on accented speech, about 6× less energy.

4. Can edda be improved by more fine-tuning on public Danish data? No.

Run (CoRal conversation validation, 1,887 clips)	WER, %	Difference vs edda, 95% CI (paired bootstrap)
edda-v0.1, unchanged	14.10	–
decoder-only fine-tune, lr 3e-6	14.24	+0.14 [–0.10, +0.39], tie
full fine-tune, lr 2e-6	14.45	+0.36 [+0.02, +0.68], slightly worse
decoder-only fine-tune, lr 1e-5	14.75	+0.65 [+0.32, +0.97], worse

10 runs completed (learning rates 1e-6 to 2e-5, conversation-only and mixed data, full vs decoder-only). Stronger settings were clearly worse (up to 34% on a 384-clip check vs 17% for edda), mostly through the audio encoder. Conclusion: better Danish lecture transcription needs new data (e.g. SDU lecture recordings), not more training on the same data. Measured fine-tuning throughput: about 250–340 hours of audio per B200 GPU-hour. The bootstrap resamples utterances, not speakers, so the intervals are likely somewhat too narrow; the conclusion (no run beats edda) does not depend on them.

5. Offline captioner (runs on a laptop)

A small script turns a recording into .srt subtitles with the models above (edda for Danish, turbo for English), splitting the audio at pauses. Tested on a MacBook Pro M4 Pro on ~7 minutes of stitched FLEURS test clips per language that were not used to tune it: Danish 9.8% WER (8.5% when the same clips are transcribed one by one, same text normaliser), English 6.8% (6.5%), both about 10× faster than real time. Not yet tested on real lecture recordings.

Method, sources and verification status

So every number can be checked. "Measured" = produced in my own runs, raw per-clip outputs and logs saved.

Method and limitations

Danish WER: public Danish ASR leaderboard harness (same normalisation for every model). English: Whisper EnglishTextNormalizer and jiwer; FLEURS full test set (647 clips), EdAcc and AMI first 2,000 test clips. Same batch size and precision for all models within a language, one model at a time per GPU. Energy, Danish: mean GPU board power while the GPU was active (nvidia-smi, 200 ms samples) divided by transcription speed. English: the GPU's own energy counter (NVML) read before and after transcription. Model loading is excluded in both (English FLEURS speed and energy include GPU warm-up; EdAcc does not); the two methods are not directly comparable, so compare energy within a language, not across. Limitations: benchmark audio, not SDU lectures; GPU energy only. GPU use: 15.2 GPU-hours for the measurement, about 35–40 in total including the fine-tuning study (see below).

Measured by me (30 September 2026; captioner test 1 October)

- All WER and energy values in tables 1, 2 and 4, the AMI repetition-loop observation, the fine-tuning results and throughput, and the captioner test in section 5.
- Sanity check: my edda-v0.1 scores equal its published model card on all four Danish sets (CoRal 15.99 / 10.52, FLEURS 7.56, FTSpeech 5.95).

Taken from published sources (checked 29–30 September 2026)

Claim	Source
Transcriber offers whisper base–large-v3, large-v3-turbo and parakeet-tdt-0.6b-v3; default is large-v3	docs.cloud.sdu.dk/Apps/transcriber.html
edda-v0.1: Apache-2.0, fine-tuned from whisper-large-v3-turbo on 2,600 h Danish; published WER	huggingface.co/danish-foundation-models/edda-v0.1
Parakeet mean WER 30.7% (not re-measured)	huggingface.co/datasets/RyeAI/danish-asr-leaderboard (published results file)
Danish evaluation harness and normalisation	github.com/Rye-A1/danish-asr-leaderboard

Test data (public; licence as listed on the dataset page)

Dataset	Licence	Source
CoRal-v3 (conversation, read-aloud test; train/val for fine-tuning)	OpenRAIL (gated access, accepted)	huggingface.co/datasets/CoRal-project/coral-v3
FLEURS (Danish, English)	CC BY 4.0	huggingface.co/datasets/google/fleurs
FTSpeech (parliament, test_balanced)	"other" on the dataset page; used for evaluation only	huggingface.co/datasets/alexandrainst/ftspeech
EdAcc (accented English)	CC BY-SA	huggingface.co/datasets/edinburghcstr/edacc
AMI meetings (ihm test)	CC BY 4.0	huggingface.co/datasets/edinburghcstr/ami

Estimated or not verified

- GPU-hours: 228.6 used before the study (read on the UCloud dashboard), 262.4 at 23:34 (read on the dashboard). 15.2 for the measurement is from a dashboard reading of 243.8 at about 19:45. The final total (about 35–40) is an estimate, since the allocation view closed at expiry.
- Energy covers the GPU only, not CPU, memory or cooling.
- Results are on benchmark audio; no SDU lecture recordings were used.
- Two further fine-tuning runs were lost to out-of-memory errors (re-run with fixed settings) and are not counted in the 10.

Code, summary tables and logs: github.com/natanaelfejes/sdu-asr-energy. Per-utterance transcripts are not shared, because the test-set licences restrict redistributing their content.